

An exact convergence criterion for the Binet factorial expansion

Candidate full solution to the question in arXiv:2102.04891

13 August 2026

Status. Candidate full answer, subject to expert review. The answer is exact but is necessarily expressed through an associated ordinary Dirichlet series. A germ-bijection below proves that arbitrary Dirichlet coefficient behavior can occur. In particular, the original Taylor radius alone cannot decide convergence.

1 Source question

The source is P. Van Mieghem, *Binet's factorial series and extensions to Laplace transforms*, arXiv:2102.04891 [1]. Section 8.4, “Open question,” on source PDF page 38 asks:

Given the Taylor series $f(t) = \sum_{k=0}^{\infty} f_k t^k$ with radius of convergence R_f , when does a factorial series (90) of the Laplace transform $\varphi(z)$ converge?

8.4 Open question

The factorial series (90) of a non-entire function may converge, whereas the Laurent series (83) does not. An example is the Binet function $\mu(z)$, whose Laurent series (83) is Stirling's famous asymptotic, but divergent series (28). Hence, the question arises: “Given the Taylor series $f(t) = \sum_{k=0}^{\infty} f_k t^k$ with radius of convergence R_f , when does a factorial series (90) of the Laplace transform $\varphi(z)$ converge?”

We can only give a partial insight. The recipe in 5 steps for a factorial series (90) of the Laplace transform $\varphi(z)$ requires that the Taylor series (89) of $g(u; 0, \beta) = f(-\beta \log(1 - u))$ around the origin $u_0 = 0$ converges within the unit circle. Its corresponding Taylor coefficient is written in terms of $f_k = \frac{1}{k!} \left. \frac{d^k f(t)}{dt^k} \right|_{t=0}$ as $g_0 = f_0$ and

$$g_m(\beta) = \frac{1}{m!} \sum_{k=1}^m k! f_k \beta^k (-1)^{m-k} S_m^{(k)} \quad \text{for } m > 0 \quad (100)$$

from which it follows that, for $m > 0$, the Taylor coefficient $g_m(\beta)$ is independent of $f_0 = f(0)$ (like any characteristic coefficient (115)). The inverse transform of (100)

$$\beta^m f_m = \frac{1}{m!} \sum_{k=1}^m k! g_k(\beta) (-1)^{m-k} S_m^{(k)} \quad \text{for } m > 0 \quad (101)$$

where $S_m^{(k)}$ is the Stirling number of the Second Kind, is deduced as in the proof of Property 5

The source fixes $\beta > 0$ and $\alpha \in \mathbb{C}$. With $w = \beta z + \alpha$, its factorial series (90) can be written

$$\mathcal{S}_{\alpha, \beta} f(z) = \beta \sum_{m=0}^{\infty} a_m \frac{m!}{\prod_{k=0}^m (w + k)} = \beta \sum_{m=0}^{\infty} a_m B(m+1, w), \quad (1)$$

where

$$G_{\alpha,\beta}(u) = (1-u)^{-\alpha} f(-\beta \operatorname{Log}(1-u)) = \sum_{m=0}^{\infty} a_m u^m \quad (2)$$

as an analytic germ at zero. Thus a_m is exactly the source coefficient $\phi_m(\alpha, \beta)$, and it is explicitly computable from the given Taylor coefficients f_k by the Stirling-number transform in the source.

2 Proof intuition

The beta factor in the m th factorial-series term is only a smoothly varying perturbation of a Dirichlet weight:

$$B(m+1, w) = \Gamma(w) m^{-w} (1 + O_w(m^{-1})).$$

The perturbing sequence has summable first differences. Multiplication by such a sequence preserves even conditional convergence, in both directions. Consequently the exact convergence question is the ordinary Dirichlet-series question for the transformed Taylor coefficients a_m .

This also yields a sharp radius trichotomy for the series in (2). A radius above one gives exponential coefficient decay; a radius below one gives exponential growth along a subsequence and hence termwise divergence; radius one is precisely the nontrivial Dirichlet case. Finally, the Binet change of variables is a bijection on analytic germs, so arbitrary Dirichlet coefficients can arise. The reduction is therefore optimal, rather than an artifact of the proof.

3 Exact criterion

All powers m^{-w} below use the real logarithm of the positive integer m .

Theorem 1 (Factorial–Dirichlet equivalence). *Fix $\beta > 0$, $\alpha \in \mathbb{C}$, and a germ f analytic at zero. Let a_m be defined by (2), let $w = \beta z + \alpha$, and suppose $\operatorname{Re} w > 0$.*

Then the factorial series (1) converges if and only if

$$\sum_{m=1}^{\infty} a_m m^{-w} \quad (3)$$

converges. The same equivalence holds for absolute convergence. The $m = 0$ term in (1) is the harmless finite term $\beta a_0/w$.

Lemma 1 (Bounded-variation multiplier). *Suppose $c_m \rightarrow c \neq 0$ and $\sum_m |c_{m+1} - c_m| < \infty$. Then $\sum b_m$ converges if and only if $\sum c_m b_m$ converges.*

Proof. If $\sum b_m$ converges, summation by parts applied to its bounded partial sums and the summable differences of c_m proves convergence of $\sum c_m b_m$. Since c_m is bounded away from zero for all sufficiently large m , the sequence $1/c_m$ also converges to $1/c$ and

$$\left| \frac{1}{c_{m+1}} - \frac{1}{c_m} \right| \leq C |c_{m+1} - c_m|.$$

The same argument with multiplier $1/c_m$ proves the converse. Finitely many initial terms do not matter. $\square$

Proof of Theorem 1. Euler's beta identity gives

$$B(m+1, w) = \frac{\Gamma(m+1)\Gamma(w)}{\Gamma(m+1+w)} = \Gamma(w)m^{-w}c_m(w), \quad c_m(w) = \frac{m^w\Gamma(m+1)}{\Gamma(m+1+w)}. \quad (4)$$

The fixed-parameter gamma-quotient expansion (equivalently, Stirling's formula with one further term) gives

$$c_m(w) = 1 + O_w(m^{-1}), \quad c_{m+1}(w) - c_m(w) = O_w(m^{-2}). \quad (5)$$

Thus $c_m(w)$ satisfies Lemma 1. Apply the lemma to $b_m = a_m m^{-w}$ in (4); multiplication by the fixed nonzero factor $\beta\Gamma(w)$ is irrelevant. This proves ordinary convergence in both directions. Moreover $|c_m(w)|$ is bounded above and away from zero eventually, which proves the absolute-convergence equivalence. $\square$

Corollary 1 (Sharp transformed-radius trichotomy). *Let R_G be the radius of convergence of (2).*

1. *If $R_G > 1$, (1) converges absolutely for every z with $\operatorname{Re}(\beta z + \alpha) > 0$.*
2. *If $R_G < 1$, (1) diverges for every such fixed z ; in fact its terms fail to tend to zero.*
3. *If $R_G = 1$, its convergence and absolute convergence regions are exactly those of (3). Hence they are governed by the usual convergence and absolute-convergence abscissae, with the usual possibility of exceptional behavior on the boundary lines.*

Proof. If $R_G > 1$, Cauchy's estimate on a circle of any radius $1 < r < R_G$ gives $|a_m| \leq C_r r^{-m}$, so (3) is absolutely convergent for every w . If $R_G < 1$, Cauchy–Hadamard gives

$$\limsup_{m \rightarrow \infty} |a_m|^{1/m} = R_G^{-1} > 1.$$

The beta factor in (4) has m th-root limit one, so the terms of (1) cannot tend to zero. The final assertion follows directly from Theorem 1 and the elementary half-plane theorem for ordinary Dirichlet series. $\square$

4 When the sum is the Laplace transform

The exact convergence statement above concerns the numerical factorial series. The following records a clean condition under which it also represents the intended Laplace transform, without concealing an interchange of a sum and an improper integral.

Theorem 2 (Absolute representation). *Suppose the chosen continuation of f along $[0, \infty)$ agrees with (2) for $0 \leq u < 1$. If $\operatorname{Re} w > 0$ and (1) converges absolutely, then the Laplace integral is absolutely convergent at z and*

$$\int_0^\infty e^{-zt} f(t) dt = \beta \sum_{m=0}^\infty a_m B(m+1, w). \quad (6)$$

Proof. By Theorem 1, absolute convergence is equivalent to $\sum |a_m| m^{-\operatorname{Re} w} < \infty$. The positive-real beta factors are comparable to these powers, so

$$\sum_{m=0}^\infty |a_m| B(m+1, \operatorname{Re} w) < \infty.$$

Since

$$B(m+1, \operatorname{Re} w) = \int_0^1 u^m (1-u)^{\operatorname{Re} w-1} du,$$

Tonelli's theorem permits absolute termwise integration. Binet's substitution $t = -\beta \log(1-u)$ then gives

$$\begin{aligned} \int_0^\infty e^{-zt} f(t) dt &= \beta \int_0^1 (1-u)^{\beta z-1} f(-\beta \log(1-u)) du \\ &= \beta \int_0^1 (1-u)^{w-1} G_{\alpha, \beta}(u) du \\ &= \beta \sum_{m=0}^\infty a_m B(m+1, w). \end{aligned}$$

The same majorant proves absolute convergence of the first integral. $\square$

No unconditional claim is made here that mere conditional convergence of (1) justifies (6); that requires an additional boundary summability hypothesis.

5 Optimality and a same-radius separation

Theorem 3 (Universality of the transformed coefficients). *For fixed $\alpha \in \mathbb{C}$ and $\beta > 0$, the transformation*

$$f(t) \mapsto (1-u)^{-\alpha} f(-\beta \operatorname{Log}(1-u))$$

is a linear bijection between analytic germs at $t = 0$ and analytic germs at $u = 0$. Its inverse is

$$f(t) = e^{-\alpha t/\beta} G(1 - e^{-t/\beta}). \quad (7)$$

Thus arbitrary power-series coefficients a_m can occur in (2); in the unrestricted generality of the source question, the Dirichlet criterion of Theorem 1 cannot be replaced by a simpler universal coefficient phenomenon.

Proof. Both substitutions are analytic near their origins and are inverse there: $u = 1 - e^{-t/\beta}$ if and only if $t = -\beta \operatorname{Log}(1-u)$. Since $(1-u)^\alpha = e^{-\alpha t/\beta}$ under this substitution, direct composition proves both inverse identities. $\square$

The next explicit pair shows more concretely that even the complete value of the original radius R_f carries too little information.

Theorem 4 (Equal R_f , opposite convergence). *Take $\alpha = 0$, $\beta = 1$, $q = 5/2$, and $R = \log q$. Define for $t \geq 0$*

$$f_A(t) = \frac{1}{q - e^{-t}}, \quad f_B(t) = \frac{1}{1 + (t/R)^2}.$$

Both Laplace transforms converge for $\operatorname{Re} z > 0$, and the Taylor series of both functions at zero has radius exactly R . Nevertheless, the factorial expansion for f_A converges absolutely for every $\operatorname{Re} z > 0$, whereas that for f_B diverges for every fixed z .

Proof. The singularities of f_A are $-\log q - 2\pi ik$, $k \in \mathbb{Z}$, so its nearest singularity is $-R$. The nearest singularities of f_B are $\pm iR$. Both Taylor radii are therefore R . On the positive real axis, f_A is bounded and $f_B(t)$ is $O(t^{-2})$, so both Laplace integrals converge when $\operatorname{Re} z > 0$.

Their transformed germs are

$$G_A(u) = f_A(-\operatorname{Log}(1-u)) = \frac{1}{q-1+u},$$

whose radius is $q-1 = 3/2 > 1$, and

$$G_B(u) = \frac{1}{1 + (-\operatorname{Log}(1-u)/R)^2}.$$

The latter has poles at $u = 1 - e^{\mp iR}$, of modulus $2 \sin(R/2)$. Since

$$e^{\pi/3} > 1 + \frac{\pi}{3} + \frac{1}{2} \left(\frac{\pi}{3}\right)^2 > \frac{5}{2},$$

we have $R = \log(5/2) < \pi/3$, and hence $2 \sin(R/2) < 2 \sin(\pi/6) = 1$. Thus G_B has radius below one. Corollary 1 gives the two opposite conclusions. $\square$

6 Scope, verification, and novelty

The theorem answers convergence at every parameter point in the source half-plane $\operatorname{Re}(\beta z + \alpha) > 0$. It also gives an absolute criterion ensuring equality with the Laplace integral. It deliberately does not infer equality from conditional convergence alone.

The gamma multiplier, both branches of the germ inverse, the two example radii, and the Tonelli step were audited independently in the accompanying verification report. No numerical computation is used as proof.

The four lightweight run indexes had no entry for this arXiv id or problem. A bounded primary-arXiv search on 13 August 2026 used the exact title and close phrases involving factorial series, Laplace transforms, Taylor coefficients, and convergence; it found the source but no later solution. The source itself already points out a classical $\beta = 1$ comparison with a Dirichlet series, citing Lauwerier and Temme. That comparison is not claimed as new. The candidate contribution is the rigorous all-parameter equivalence including conditional convergence, the sharp transformed-radius trichotomy, the germ-universality/optimality theorem, and the equal- R_f separation. Novelty confidence is therefore moderate rather than high pending a specialist check of the classical factorial-series literature.

Human-review recommendation. Review as a candidate full answer. The highest-value checks are Theorem 1's two-way conditional-convergence argument and whether an equivalent universality/radius formulation already occurs in the older literature cited by the source.

References

- [1] P. Van Mieghem, *Binet's factorial series and extensions to Laplace transforms*, arXiv:2102.04891v7 (2023).